JINCHEN HE

Fermi National Accelerator Laboratory, Batavia ◇ Illinois 60510, USA
(+1) 240-413-9989 ◇ jinchen@terpmail.umd.edu ◇ INSPIRE HEP ◇ Personal Page

EDUCATION

University of Maryland, College Park *September 2022 - August 2026*
Doctor of Philosophy (Ph.D.) in Physics [Ph.D. Thesis]

University of California, Berkeley *January 2020 - June 2020*
Exchange student in Physics

University of Chinese Academy of Sciences *September 2017 - June 2021*
Bachelor of Science (B.S.) in Physics

APPOINTMENTS

Fermi National Accelerator Laboratory *August 2026 - August 2029*
Fixed Term Full time Research Associate

Argonne National Laboratory *August 2024 - July 2026*
Research Assistant

University of Maryland *September 2022 - July 2026*
Graduate Assistant

Shanghai Jiao Tong University *July 2021 - July 2022*
Research Assistant

HONORS AND AWARDS

Theory session Award for New Perspectives at Fermilab	*2025*
APS GHP Student Travel Grant	*2025*
Ralph Myers & Friends of Physics Award, Outstanding Teaching Assistant	*2023*
PITT PACC travel award for Pheno 2023	*2023*
Dean's Fellowship at University of Maryland	*2022*
Second Award of 2021 in Lattice Parton Collaboration	*2021*
Academic Scholarship at University of Chinese Academy of Sciences	*2017 - 2019*
Merit Student at University of Chinese Academy of Sciences	*2017*
Excellent Student Cadre at University of Chinese Academy of Sciences	*2017*

GRANTS AND PROPOSALS

The 3D structure of nucleon from lattice QCD *Jan. 2026 - Jan. 2027*
Principal Investigator (PI) *NERSC*

· Computed nucleon transverse momentum-dependent parton distributions (TMDs) from lattice QCD within the Large Momentum Effective Theory (LaMET) framework.
· 200,000 Nvidia A100 GPU hours of Perlmutter@NERSC.

Learning Field Transformations to Accelerate Hybrid Monte Carlo *June 2025 - Jan. 2027*
Principal Investigator (PI) *ALCF*

· We propose to accelerate hybrid Monte Carlo simulations of lattice gauge fields via a neural network-constructed field transformation. We first apply this approach to the 2D U(1) gauge theory, and then plan to extend it to the SU(3) non-abelian case.

· 24,000 Nvidia A100 GPU hours of Sophia@ALCF.

3D Imaging of the Pion on a Fine Lattice *June 2024 - June 2026*
Principal Investigator (PI) *USQCD*

· We propose to calculate the transverse-momentum-dependent distribution (TMD) of the pion valence quark using staggered HISQ gauge configurations with clover fermions on the hypercubic (HYP) gauge background at the pion mass $m_\pi = 300$ MeV with fine spacing $a = 0.06$ fm.
· 130,000 Nvidia A100 GPU hours of LQ2@FNAL.

PROFESSIONAL SERVICE

Moderator of "AI for HEP" community on alphaXiv *April 2025 - October 2025*

· Manage the "AI for HEP" community on alphaXiv, including coordinating online discussions and curating topics at the intersection of high-energy physics and AI.
· Co-host online "AI for HEP" seminars for the community every two to three weeks, invite researchers to present their work, and facilitating interactive Q&A sessions.

MENTORING

Shivam Panda *April 2026 - Now*
Undergraduate Student, Oxford

· Mentored Shivam on applying neural networks to lattice gauge theory.

Justin Dean *February 2024 - Now*
Undergraduate Student, Virginia Tech

· Guided Justin on a project studying the effects of Gribov copies in lattice calculations of gluon correlators.

Qi Cai *February 2022 - July 2022*
Undergraduate Student, Shanghai Jiao Tong University

· Provided guidance to Qi on foundational lattice gauge theory, covering core concepts of lattice QCD as well as the computation and analysis of two-point correlation functions.

PUBLICATIONS

Lead-author publications

· X. Gao, **J. He**, J. Lin, S. Mukherjee, P. Petreczky, R. Zhang and Y. Zhao, "*Nucleon parton distribution functions from boosted correlations in the Coulomb gauge*," JHEP 10 (2026) 041 arXiv:2602.11283 [hep-lat].
· D. Bollweg, X. Gao, **J. He**, S. Mukherjee and Y. Zhao (Apr. 2025). "*Transverse-momentum-dependent pion structures from lattice QCD: Collins-Soper kernel, soft factor, TMDWF, and TMDPDF*", Phys. Rev. D 112, (2025) 3 arXiv:2504.04625 [hep-lat]
· X. Gao, **J. He**, R. Zhang and Y. Zhao (Aug. 2024). "*Systematic Uncertainties from Gribov Copies in Lattice Calculation of Parton Distributions in the Coulomb gauge*", Chin.Phys.Lett. 41 (2024) 12 arXiv:2408.05910 [hep-lat]
· **J. C. He** *et al.* (Lattice Parton Collaboration) (Nov. 2022). "*Unpolarized Transverse-Momentum-Dependent Parton Distributions of the Nucleon from Lattice QCD*", Phys.Rev.D 109 (2024) 11 arXiv:2211.02340 [hep-lat]
· **J. He**, D. A. Brantley, C. C. Chang, *et al.* (CalLat Collaboration) (Apr. 2021). "*Detailed analysis of excited state systematics in a lattice QCD calculation of* g_A", Phys.Rev.C 105 (2022) 6 arXiv:2104.05226 [hep-lat]

- **J. He**, X. Y. Jin, J. C. Osborn and Y. Zhao (Aug. 2025). "*Neural Field Transformations for Hybrid Monte Carlo: Architectural Design and Scaling*", NeurIPS 2025 Workshop on Machine Learning for Physical Sciences arXiv:2511.02018 [hep-lat]
- **J. He**, X. Jiang, F. Yao and D.-J. Zhao (Sep. 2026). "*LaMET-Agent: An Agent Framework for Large-Momentum Effective Theory Analysis*", arXiv:2609.32225 [hep-lat].
- **[In preparation]** "*Unpolarized, Helicity, and Transversity Proton TMDPDFs from Lattice QCD: Spin Dependence of the Nonperturbative Transverse Structure*"

Key-contributor publications

- J. W. Chen, X. Gao, **J. He**, *et al.* (May 2025). "*LaMET's Asymptotic Extrapolation vs. Inverse Problem*", arXiv:2505.14619 [hep-lat]
- H. Liu, **J. He**, L. Liu, *et al.* (Jul. 2022). "*Hidden-charm Hexaquarks from Lattice QCD*", Sci.China Phys.Mech.Astron. 67 (2024) 1 arXiv:2207.00183 [hep-lat]
- J. Hua, M. H. Chu, **J. C. He**, *et al.* (Lattice Parton Collaboration) (Jan. 2022). "*Pion and Kaon Distribution Amplitudes from Lattice QCD*", Phys.Rev.Lett. 129 (2022) 13 arXiv:2201.09173 [hep-lat]

Contributor publications

- X. Gao, A. Hanlon, **J. He**, X. Jiang, S. Mukherjee, P. Petreczky, Q. Shi and Y. Zhao (Nov. 2026). "*Unpolarized proton PDF with physical quark masses and large momenta*," arXiv:2611.12730 [hep-lat].
- M. Zhu, M. Tian, X. Yang, T. Zhou, P. Zhu, E. Chertkov, S. Liu, Y. Du, L. Yuan and **J. He**, *et al.* "*Probing the Critical Point (CritPt) of AI Reasoning: a Frontier Physics Research Benchmark*", arXiv:2509.26574 [cs.AI].
- X. Gao, **J. He**, Y. Su, R. Zhang and Y. Zhao (Aug. 2024). "*Comments on 'Non-local Nucleon Matrix Elements in the Rest Frame'*", arXiv:2408.04674 [hep-lat]
- M. H. Chu, **J. C. He**, J. Hua, *et al.* (Lattice Parton Collaboration) (Jun. 2023). "*Lattice calculation of the intrinsic soft function and the Collins-Soper kernel*", JHEP 08 (2023) 172 arXiv:2306.06488 [hep-lat]
- M. H. Chu, **J. C. He**, J. Hua, *et al.* (Lattice Parton Collaboration) (Feb. 2023). "*Transverse-Momentum-Dependent Wave Functions of Pion from Lattice QCD*", Phys.Rev.D 109 (2024) 9 arXiv:2302.09961 [hep-lat]

Open-Source contributions

- `LaMETLat`: Python package for lattice calculations in Large Momentum Effective Theory. Publicly available at github.com/Greyyy-HJC/LaMETLat.

SEMINARS AND CONFERENCE PRESENTATIONS

Seminars

- "3D Imaging of the Pion on a Fine Lattice", Student Lunch Talk, Argonne National Laboratory, April 15, 2026.
- "3D Imaging of the Pion on a Fine Lattice", Rising Researcher Seminar, Institute for Nuclear Theory (INT), March 10, 2026.
- "3D Imaging of the Pion on a Fine Lattice", Special CFNS seminar, Stony Brook University, October 15, 2025.
- "Effective field theory for positronium in relativistic motion", Nuclear Theory Seminar, University of Maryland, April 17, 2025.
- "NRQED: Lamb Shift and Relativistic Hydrogen Atom", Theory Seminar, Argonne, January 12, 2024.
- "Unpolarized Nucleon TMDPDFs from LQCD", EFT Seminar, Technische Universität München, November 24, 2023.

Invited Talks

- "3D Imaging of the Pion on a Fine Lattice", 15th Conference on the Intersections of Particle and Nuclear Physics, University of Wisconsin–Madison, June 12, 2025.
- "Imaging of the Pion on a Fine Lattice", USQCD All-Hands Meeting, Online, April 19, 2024.

- "Unpolarized Nucleon TMDPDFs from LQCD", TMDs: Towards a Synergy between Lattice QCD and Global Analyses, Stony Brook University, June 22, 2023.

Contributed Talks

- "Proton TMDPDFs from Lattice QCD: Spin Dependence of the Non-perturbative Transverse Structure", Lattice 2026, University of Maryland, College Park, MD, U.S., July 28, 2026.
- "Proton TMDPDFs from Lattice QCD: Spin Dependence of the Non-perturbative Transverse Structure", Division of Particles and Fields (DPF) 2026, Fermilab, Batavia, IL, U.S., July 22, 2026.
- "Proton TMDPDFs from Lattice QCD: Spin Dependence of the Non-perturbative Transverse Structure", 2026 Meeting on Lattice Parton Physics from Large Momentum Effective Theory (LaMET 2026), Jagiellonian University, Cracow, Poland (Online), July 8, 2026.
- "Nucleon Parton Distribution Functions from Boosted Correlators in CG", The 36th Midwest Theory Get Together, Argonne, September 26, 2025.
- "Nucleon Parton Distribution Functions from Boosted Correlators in CG", QGT Topical Collaboration Meeting, Argonne, September 19, 2025.
- "3D Imaging of the Pion on a Fine Lattice", New Perspectives 2025, Fermilab, July 15, 2025.
- "Nucleon Parton Distribution Functions from Boosted Correlators in CG", APS Topical Group on Hadronic Physics, Anaheim, March 16, 2025.
- "Effective Field Theory for Positronium in Relativistic Motion", Midwest Theory Get Together, Argonne, October 18, 2024.
- "Systematic Uncertainties from Gribov Copies in CG-Fixed Correlators", QGT Topical Collaboration Meeting, Temple University, September 13, 2024.
- "Systematic Uncertainties from Gribov Copies in Lattice Calculation of Quasi-distributions in the Coulomb gauge", LaMET 2024, University of Maryland, August 14, 2024.
- "Nucleon TMDPDFs from Lattice QCD", Phenomenology Symposium 2023, University of Pittsburgh, May 8, 2023.
- "Unpolarized Transverse-Momentum-Dependent PDF from Lattice QCD", LaMET 2022, Argonne, December 2, 2022.
- "Pion and Kaon Distribution Amplitudes with LaMET", LaMET 2021, Online, December 8, 2021.
- "Distribution amplitudes from lattice QCD with LaMET", International Workshop on Heavy Quark Physics, Online, November 24, 2021.

Poster Presentation

- "Mapping the 3D Parton Structures of Hadrons with Lattice QCD", 2026 SciDAC Principal Investigator (PI) Meeting, Hilton Washington DC/Rockville Hotel & Executive Meeting Center, USA, August 4–6, 2026 (meeting dates).
- "Nucleon Parton Distribution Functions from Boosted Correlators in CG", Physics Opportunities at an Electron-Ion Collider XI, Florida International University, February 25, 2025.

SKILLS

Programming & Software	Python, C++, Linux, Git, Mathematica, LaTeX, QUDA
Machine Learning & Data Science	PyTorch, NumPy, SciPy, Pandas, Matplotlib
High-Performance Computing	Experience with large-scale GPU/CPU clusters: OLCF (Andes/Frontier, NERSC), ALCF (Polaris/Sophia), FNAL (LQ2), LCRC (Swing/Improv)
Languages	Chinese (Native), English (Professional Working Proficiency)